Operation Dim Star: Exoplanet Detection, Orbital Modeling, and Habitability Analysis

Alliance Astronomical Analysis Team, *Rebel Alliance Signal Intelligence*

***Abstract*—Photometric light curve data obtained from stars STAC-1, STAC-2, and STAC-3 were systematically analyzed to identify transiting exoplanets and evaluate candidates for a potential Rebel base location. Periodic flux decreases were detected in all three systems. System STAC-1 hosts a Neptune-sized world ($R_p = 5.98\,R_\oplus$) on a 4.0-day orbit at 0.0493 AU. System STAC-2 harbors a Super-Earth ($R_p = 2.76\,R_\oplus$) orbiting at 0.3373 AU with an 80.0-day period. System STAC-3 hosts a Super-Earth candidate ($R_p = 1.64\,R_\oplus$) with a 40.0-day period at 0.1817 AU. Comparative habitability evaluation based on stellar luminosity ($0.08\,L_\odot$) places STAC-3b directly adjacent to the circumstellar habitable zone ($d_{\text{HZ}} = 0.283$ AU). STAC-3b is determined to be the most promising candidate for detailed atmospheric characterization and base deployment.**

***Index Terms*—Exoplanets, Transit Photometry, Kepler's Third Law, Astrobiology, Habitable Zone, STAC WARS.**

I. Introduction

FOLLOWING the destruction of the Death Star, the Rebel Alliance requires an unchartered terrestrial body for relocation. Photometric monitoring of three candidate host stars—STAC-1, STAC-2, and STAC-3—yielded light curves indicating periodic dips in stellar flux. This paper documents the analytical pipeline used to confirm planetary transits, compute physical and orbital parameters, and identify the optimal target world.

II. Transit Detection and Periodicity

A. Detection Mechanism

When an exoplanet's orbital plane aligns with the line of sight, the planet periodically blocks a fraction of the stellar disk, producing a periodic dip in relative brightness:

$$\delta = \frac{F_{\text{normal}} - F_{\text{transit}}}{F_{\text{normal}}} \tag{1}$$

where δ is the transit depth, F_{normal} is the baseline continuum flux, and F_{transit} is the flux recorded during transit.

B. Light Curve Characterization

1) STAC-1: Light curve observations demonstrate periodic flux reductions to $F = 0.9970$ at $t = 3, 8, 12, 16,$ and 20 days. The recurrence interval stabilizes at $\Delta t = 4.0$ days.

$$P_1 = 4.0 \text{ days} = 0.01095 \text{ years}$$
$$\delta_1 = \frac{1.0000 - 0.9970}{1.0000} = 0.0030 \quad (0.30\%)$$

Submission for STAC WARS Exoplanet Detection Challenge.

2) STAC-2: Brightnesses drop to $F = 0.9990$ at $t = 20, 100,$ and 180 days, yielding a uniform orbital period of 80.0 days.

$$P_2 = 80.0 \text{ days} = 0.21903 \text{ years}$$
$$\delta_2 = \frac{1.0000 - 0.9990}{1.0000} = 0.0010 \quad (0.10\%)$$

3) STAC-3: Periodic dips occur at intervals of 40.0 days ($t = 20, 60, 100, 140, 170$ days), with an average transit brightness of $F = 0.9991$.

$$P_3 = 40.0 \text{ days} = 0.10951 \text{ years}$$
$$\delta_3 = \frac{1.0000 - 0.9991}{1.0000} = 0.0009 \quad (0.09\%)$$

III. Planetary Radius Derivation

Assuming a spherical planet transiting a spherical star, the geometric transit depth relates directly to the ratio of radii:

$$\delta \approx \left(\frac{R_p}{R_s}\right)^2 \implies R_p = R_s\sqrt{\delta} \tag{2}$$

Using standard physical constants ($R_\odot = 696,000$ km, $R_\oplus = 6,371$ km):

A. STAC-1b

For $R_{s1} = 1.00\,R_\odot = 696,000$ km:

$$R_{p1} = 696,000 \times \sqrt{0.0030} = 38,121.5 \text{ km}$$
$$R_{p1} = \frac{38,121.5}{6,371} = 5.98\,R_\oplus \quad \text{(Neptune-sized)}$$

B. STAC-2b

For $R_{s2} = 0.80\,R_\odot = 556,800$ km:

$$R_{p2} = 556,800 \times \sqrt{0.0010} = 17,607.7 \text{ km}$$
$$R_{p2} = \frac{17,607.7}{6,371} = 2.76\,R_\oplus \quad \text{(Super-Earth)}$$

C. STAC-3b

For $R_{s3} = 0.50\,R_\odot = 348,000$ km:

$$R_{p3} = 348,000 \times \sqrt{0.0009} = 10,440.0 \text{ km}$$
$$R_{p3} = \frac{10,440.0}{6,371} = 1.64\,R_\oplus \quad \text{(Super-Earth)}$$

IV. Orbital Dynamics

Semi-major axes (a) are calculated using Kepler's Third Law in solar system units:

$$P^2 = \frac{a^3}{M} \implies a = \sqrt[3]{P^2 \cdot M} \tag{3}$$

where P is in years, M in solar masses ($M_\odot$), and a in astronomical units (AU). Unit conversion uses $1\ \text{AU} = 1.496 \times 10^8$ km.

- **STAC-1b** ($M = 1.00\,M_\odot, P = 0.01095$ yr):
  $a_1 = \sqrt[3]{(0.01095)^2 \times 1.00} = 0.0493$ AU $= 7.38{\times}10^6$ km
- **STAC-2b** ($M = 0.80\,M_\odot, P = 0.21903$ yr):
  $a_2 = \sqrt[3]{(0.21903)^2 \times 0.80} = 0.3373$ AU $= 5.05{\times}10^7$ km
- **STAC-3b** ($M = 0.50\,M_\odot, P = 0.10951$ yr):
  $a_3 = \sqrt[3]{(0.10951)^2 \times 0.50} = 0.1817$ AU $= 2.72{\times}10^7$ km

V. Candidate Evaluation and Selection

A. Habitable Zone Determination

The habitable zone distance (d_{HZ}) scales with stellar luminosity:

$$d_{\text{HZ}} \approx \sqrt{\frac{L}{L_\odot}}\ \text{AU} \tag{4}$$

- **STAC-1:** $L_1 = 1.00\,L_\odot \implies d_{\text{HZ1}} = 1.000$ AU.
- **STAC-2:** $L_2 = 0.40\,L_\odot \implies d_{\text{HZ2}} = 0.632$ AU.
- **STAC-3:** $L_3 = 0.08\,L_\odot \implies d_{\text{HZ3}} = 0.283$ AU.

B. Target Selection Rationale

As detailed in Table I, **STAC-3b** is identified as the prime candidate for further study based on three key parameters:

1. **Surface Density Profile:** At $1.64\,R_\oplus$, STAC-3b falls below the empirical $1.8 - 2.0\,R_\oplus$ threshold dividing rocky terrestrial worlds from volatile-rich mini-Neptunes. 2. **Thermal Regime:** STAC-1b (0.0493 AU) is severely scorched, while STAC-2b (0.3373 AU) orbits at nearly half its habitable distance (0.632 AU). STAC-3b (0.1817 AU) sits closest to its system's habitable zone boundary (0.283 AU). 3. **Host Star Stability:** STAC-3 is a low-luminosity dwarf ($0.08\,L_\odot, 3800$ K), reducing overall flare hazards relative to high-mass stellar hosts.

C. Proposed Follow-up Observation

We recommend conducting **Transmission Spectroscopy** during transit using space-based spectroscopic sensors. This will measure atmospheric molecular absorption lines (H_2O, CO_2, O_3, CH_4) to evaluate atmospheric composition, pressure, and biological indicators prior to ground operation deployment.

VI. Conclusion

STAC-3b represents the optimal balance of terrestrial planet radius, moderate orbital equilibrium, and stellar energy environment. Immediate follow-up spectrographic profiling is recommended to confirm atmospheric suitability for the Alliance base.

TABLE I
COMPREHENSIVE PHYSICAL, ORBITAL, AND HABITABILITY SUMMARY

System	**Period P (days)**	**Transit Depth δ**	**Radius R_p ($R_\oplus$)**	**Semi-major axis a (AU)**	**$d_{\mathbf{HZ}}$ (AU)**	**Classification**
STAC-1b	4.0	0.0030	5.98	0.0493	1.000	Neptune-sized
STAC-2b	80.0	0.0010	2.76	0.3373	0.632	Super-Earth
STAC-3b	**40.0**	**0.0009**	**1.64**	**0.1817**	**0.283**	**Super-Earth (Selected)**